Title: Extreme Value Theory for Portfolio Risk & Stress Testing

Enhancing Tail Risk Modeling Beyond Traditional Methods

1. Introduction

Financial markets are characterized by extreme events that are not adequately captured by traditional risk models. Standard approaches such as Value at Risk (VaR), particularly under normality assumptions, tend to underestimate the likelihood and magnitude of large losses.

This project extends a previously developed stress testing framework (based on historical simulation and Monte Carlo methods) by applying **Extreme Value Theory (EVT)** to explicitly model tail risk.

Two EVT approaches are implemented:

- **Generalized Extreme Value (GEV)** — Block Maxima method
- **Generalized Pareto Distribution (GPD)** — Peaks Over Threshold (POT) method

The objective is to improve estimation of extreme losses and strengthen portfolio stress testing under rare market events.

Portfolio Description

The analysis is conducted on a diversified equity portfolio consisting of:

- Apple Inc. (AAPL)
- Tesla Inc. (TSLA)
- JPMorgan Chase & Co. (JPM)
- Vanguard FTSE Emerging Markets ETF (VWO)

Portfolio value is set at:

$1,000,000 USD (arbitrary benchmark capital)

All risk measures are interpreted in USD terms.

2. Theoretical Background

2.1 Generalized Extreme Value (GEV)

The GEV distribution models the distribution of block maxima:

- Captures extreme values over fixed intervals (e.g., weekly losses)
- Characterized by:
    - Location **(μ)** parameter
    - Scale **(σ)** parameter
    - Shape **(ξ)** parameter

The shape parameter determines tail behavior:

- $\xi > 0$ → Heavy tail (Fréchet)
- $\xi = 0$ → Light tail (Gumbel)
- $\xi < 0$ → Bounded tail (Weibull)

2.2 Generalized Pareto Distribution (GPD)

The GPD models exceedances over a high threshold:

- Uses more extreme observations than GEV
- Provides more efficient estimation of extreme losses

Key advantage:

The Peaks Over Threshold method utilizes all extreme observations rather than a single maximum per period.

3. Data and Methodology

3.1 Data Description

Portfolio returns aggregated from multi-asset holdings.

Losses defined as:

$$\textbf{Loss} = -\textbf{Return}$$

Focus: extreme downside movements (tail events)

4. Extreme Value Theory (GEV) – Block Maxima Approach

4.1 Methodology

To model extreme downside risk under the Block Maxima framework, the portfolio losses were aggregated at a monthly frequency. The minimum monthly returns were extracted and transformed into losses:

$$\boldsymbol{L_t = -\min\left(R_{t,\text{monthly}}\right)}$$

This transformation ensures that the analysis focuses on extreme negative realizations in each period.

A Generalized Extreme Value (GEV) distribution was then fitted to the monthly loss series using Maximum Likelihood Estimation.

4.2 Empirical Characteristics

The monthly loss distribution exhibits strong evidence of non-normality and heavy tails:

- **Mean loss: 0.0325**
- **Standard deviation: 0.0216**
- **Maximum loss: 0.1585**
- **Kurtosis: 17.18 (extremely fat-tailed behavior)**

These statistics confirm that extreme losses occur far more frequently than would be expected under Gaussian assumptions.

4.3 GEV Parameter Estimates

The fitted GEV model produced the following estimates:

$$\boldsymbol{\mu = 0.02330, \sigma = 0.01117, \xi = 0.19268}$$

The positive shape parameter indicates that the distribution belongs to the *Fréchet domain of attraction*, confirming the presence of heavy tails in the extreme loss process.

4.4 GEV-Based Risk Measures

Using the fitted model, tail risk measures were computed:

Measure	Value
VaR (95%)	6.61%
VaR (99%)	9.96%
ES (95%)	15.85%

In monetary terms (portfolio value = 1,000,000$):

VaR (95%): 66,084$

VaR (99%): 99,625$

ES (95%): 158,547$

The gap between VaR and ES confirms substantial tail severity, where extreme losses beyond the VaR threshold are significantly larger on average.

4.5 Model Interpretation

The GEV results highlight three key insights:

1. Heavy-tailed behavior is present in extreme losses, confirmed by a positive shape parameter.
2. Risk is asymmetric, with extreme downside events dominating portfolio behavior.
3. Expected Shortfall significantly exceeds VaR, reinforcing that VaR alone is insufficient for capturing tail risk.

4.6 Diagnostics

- Histogram vs GPD density
- Empirical vs theoretical CDF
- QQ plot
- Kolmogorov-Smirnov test

5. Extreme Value Theory (EVT) – Peaks Over Threshold (POT) Model

5.1 Methodology

Threshold selected at:

$$\boldsymbol{u = 95\text{th percentile of losses}}$$

Exceedances:

$$\boldsymbol{W = X - u \text{ for } X > u}$$

*GPD fitted using MLE***.**

To model tail risk more accurately, the Peaks Over Threshold (POT) approach was applied to the portfolio's monthly loss distribution. A threshold corresponding to the 95th percentile of losses was selected, ensuring focus on extreme observations while maintaining sufficient sample size for stable estimation.

5.2 Tail Behavior and Parameter Estimates

Maximum Likelihood Estimation of the Generalized Pareto Distribution (GPD) produced the following parameter estimates:

$$\boldsymbol{\hat{\sigma} = 0.02254, \qquad \hat{\xi} = 0.31168}$$

The positive shape parameter indicates that the portfolio exhibits a heavy-tailed (Fréchet-type) distribution. This confirms that extreme losses decay slowly and are more likely than under a normal distribution assumption.

5.3 Risk Measures

The estimated tail risk measures are:

Measure	Value
VaR (99%)	11.76%
VaR (99.5%)	14.69%
ES (99%)	17.25%

In monetary terms (portfolio value = 1,000,000):

VaR (99%): 117,645$

VaR (99.5%): 146,889$

ES (99%): 172,460 $

The gap between VaR and ES highlights significant tail severity, confirming that losses beyond the VaR threshold are substantially larger on average.

5.4 Model Validation

Model fit and adequacy was assessed using:

- **Histogram fit of exceedances**
- **Empirical vs theoretical CDF comparison**
- **QQ plot of tail quantiles -** QQ plots show minor deviation in extreme tail (expected in EVT)
- **Kolmogorov–Smirnov test**

The KS test returned:

- **D = 0.0426**
- **p-value = 0.989**

indicating no statistically significant deviation between the fitted GPD and empirical exceedances.

However, this result should be interpreted cautiously due to the small number of exceedances, which limits the power of formal goodness-of-fit tests in extreme value settings.

5.5 Back testing Performance

Violation rate calculated as:

$$\textbf{Violation Rate} = \frac{\textbf{Losses > VaR}}{\textbf{Total Observations}}$$

The model produced a violation rate of approximately:

$$\textbf{Violation Rate} = \mathbf{0.0132}$$

This is close to the expected 1% level for a 99% VaR model, indicating strong predictive performance in tail estimation.

Key Insights

- Financial returns exhibit **fat tails**
- EVT provides a **better framework for extreme risk**
- POT (GPD) is more efficient than Block Maxima (GEV)

5.6 Interpretation

The results confirm that:

- **Portfolio returns exhibit significant heavy-tailed behavior**
- **Extreme losses are more severe than predicted by Gaussian models**
- **Tail risk is material and cannot be ignored in risk management**
- **Expected Shortfall provides a more realistic measure of extreme loss than VaR alone**

Overall, the POT-GPD framework provides a robust and theoretically consistent approach to modeling extreme portfolio risk.

Value at Risk (VaR) Formula

$$\mathrm{VaR}_p = u + \frac{\sigma}{\xi}\left[\left(\frac{N}{N_u}(1-p)\right)^{-\xi} - 1\right]$$

Expected Shortfall (ES) Formula

$$ES_p = \frac{\mathrm{VaR}_p + \sigma - \xi u}{1-\xi}$$

6. GEV vs GPD – Extreme Risk Comparison

To ensure robustness, both Block Maxima (GEV) and Peaks Over Threshold (GPD) frameworks were applied. This allows comparison of tail risk estimates under two different EVT paradigms.

6.1 Parameter Comparison

Model	Parameter	Estimate
GEV	Shape (ξ)	0.1927
GPD	Shape (ξ)	0.3117

Interpretation:

- Both models indicate **positive shape parameters**
- GPD suggests **heavier tails than GEV**
- This is expected since POT uses more extreme observations

6.2 Value-at-Risk Comparison

Confidence Level	GEV VaR	GPD VaR
95%	~6.61%	~11.76%
99%	~9.96%	~11.76–14.69%

Key Insight:

- GPD produces **more conservative (higher) tail risk estimates**
- GEV smooths extremes due to block aggregation
- POT model captures more granular tail behavior

6.3 Expected Shortfall Comparison

Model	ES Estimate
GEV	~15.85%
GPD	~17.25%

Interpretation:

- Both models confirm **ES > VaR**
- GPD produces higher ES due to direct modeling of exceedances
- This highlights **model risk sensitivity in extreme quantiles**

GEV vs GPD Comparison

Feature	GEV	GPD
Data used	Block maxima	Tail exceedances
Shape ξ	0.1927	0.3117
Tail sensitivity	Moderate	High
VaR magnitude	Lower	Higher
Stability	Smoother	More responsive

6.4 Key Findings

- **Portfolio returns are non-normal and heavy-tailed**
- **Extreme losses dominate risk profile**
- **VaR significantly underestimates tail severity**
- **Expected Shortfall is more reliable for extreme risk**
- **GPD provides most realistic stress estimates**

6.5 Key Takeaways from EVT Comparison

- Both GEV and GPD confirm **strong heavy-tailed behavior**
- GPD is more sensitive and produces **higher tail risk estimates**
- GEV provides smoother long-run extreme behavior
- Differences highlight **model risk in extreme value estimation**
- Using both approaches improves robustness of risk assessment

6.6 Final Interpretation

The consistency between GEV and GPD results strengthens the conclusion that the portfolio exhibits structurally heavy-tailed risk behavior. However, the magnitude of risk estimates varies across models, reinforcing the importance of using multiple EVT frameworks when assessing extreme financial risk.

7. Stress Testing Extension (EVT-Based)

To extend beyond static risk measures, EVT is used to simulate extreme loss scenarios.

Scenarios:

- Moderate (95%)
- Severe (99%)
- Extreme (99.9%)

Insight:

EVT enables forward-looking stress testing by simulating losses beyond historical observations.

8. Macroeconomic Risk Extension: FX & Capital Flows

Recent developments involving *dividend repatriation* by multinational firms listed on the Nairobi Securities Exchange (**NSE**) highlight the impact of capital outflows on currency stability.

Such events increase demand for foreign currency, potentially exerting depreciation pressure on the domestic currency.

This motivates the inclusion of FX-driven stress scenarios, where currency movements amplify portfolio losses.

9. Link to Previous Work & Model Improvement

This project builds directly on a prior stress testing framework, which relied on historical simulation and Monte Carlo methods.

A key limitation identified in that analysis was the assumption of normally distributed returns, which led to a systematic underestimation of extreme losses.

The current EVT-based framework addresses this limitation by explicitly modeling the tail of the loss distribution using:

- Generalized Extreme Value **(GEV)**
- Generalized Pareto Distribution **(GPD)**

This allows for:

- more accurate estimation of rare events
- improved modeling of heavy-tailed behavior
- enhanced stress testing capability

This work serves as a foundation for extending stress testing into macro-driven scenarios such as foreign exchange shocks and systemic crises.

10. Conclusion

Extreme Value Theory provides a robust framework for modeling financial tail risk.

Key findings:

- Financial returns exhibit heavy tails
- Traditional models underestimate extreme losses
- GPD offers more efficient tail estimation than GEV
- EVT enhances stress testing capabilities

11. Future Work

- Apply EVT to FX markets (USD/KES)
- Incorporate macroeconomic shocks
- Develop integrated stress testing frameworks